

OBJECT DETECTION MODEL

Presented by Ragbooms Group

OVERVIEW

The Object Detection System for Traffic Management uses advanced machine learning to identify and track traffic-related objects like vehicles, pedestrians, and signs in real-time. It improves road safety by detecting violations such as speeding or red-light running. The system also optimizes traffic flow by analyzing congestion and adjusting signals accordingly. By automating enforcement and logging violations, it supports law enforcement efforts. Overall, it enhances traffic management efficiency, contributing to smarter, safer cities.

OBJECTIVE

- **Enhance Traffic Safety** : Detect hazards like pedestrians and violations.
- **Optimize Traffic Flow** : Monitor and reduce congestion in real-time.
- **Automate Violation Detection** : Automatically log violations like speeding and illegal parking.
- **Real-Time Insights** : Provide immediate data for quick decision-making.
- **Seamless Integration** : Integrate with existing traffic systems for better coordination.

FEATURE OF THE SYSTEM

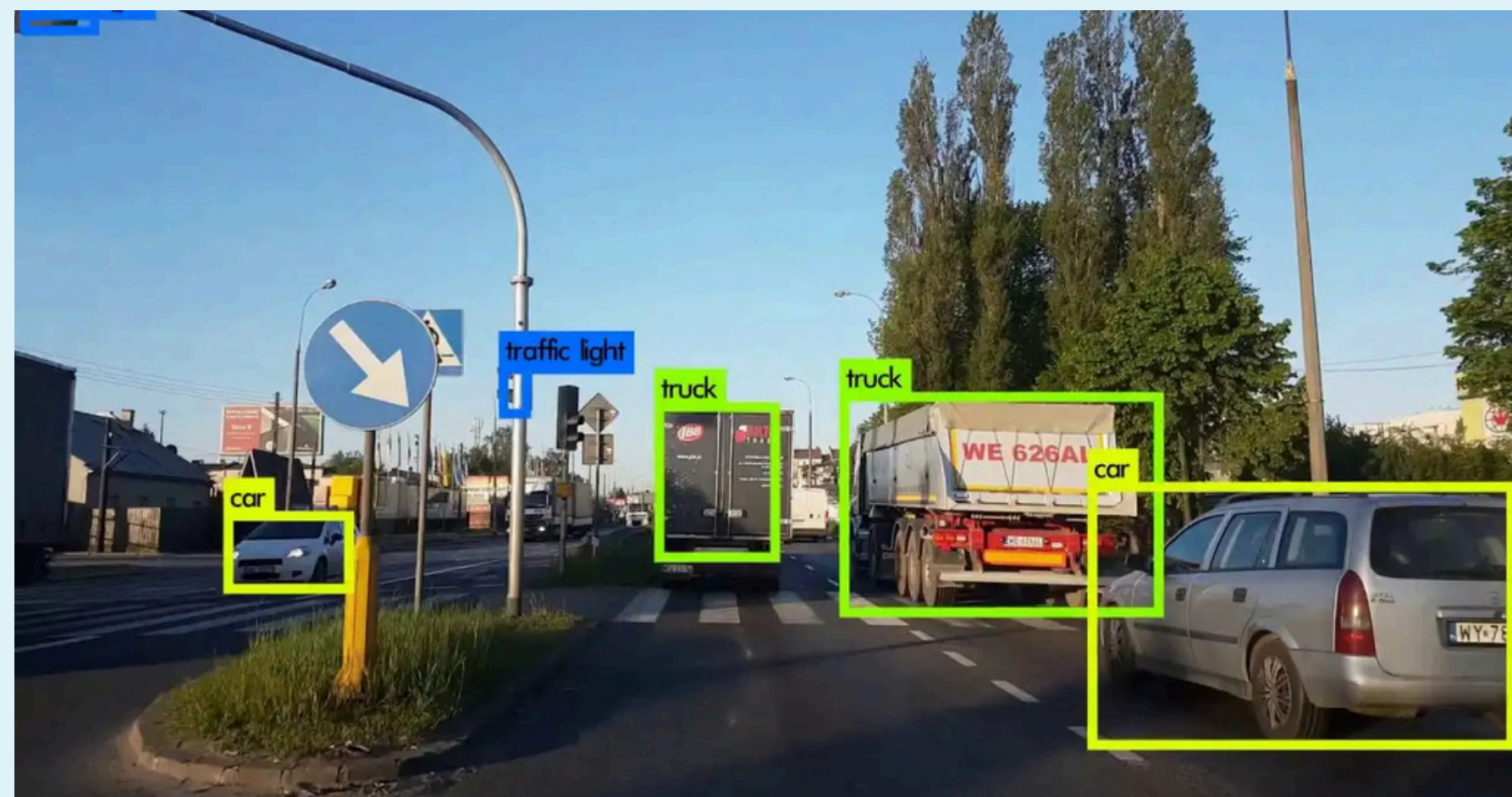
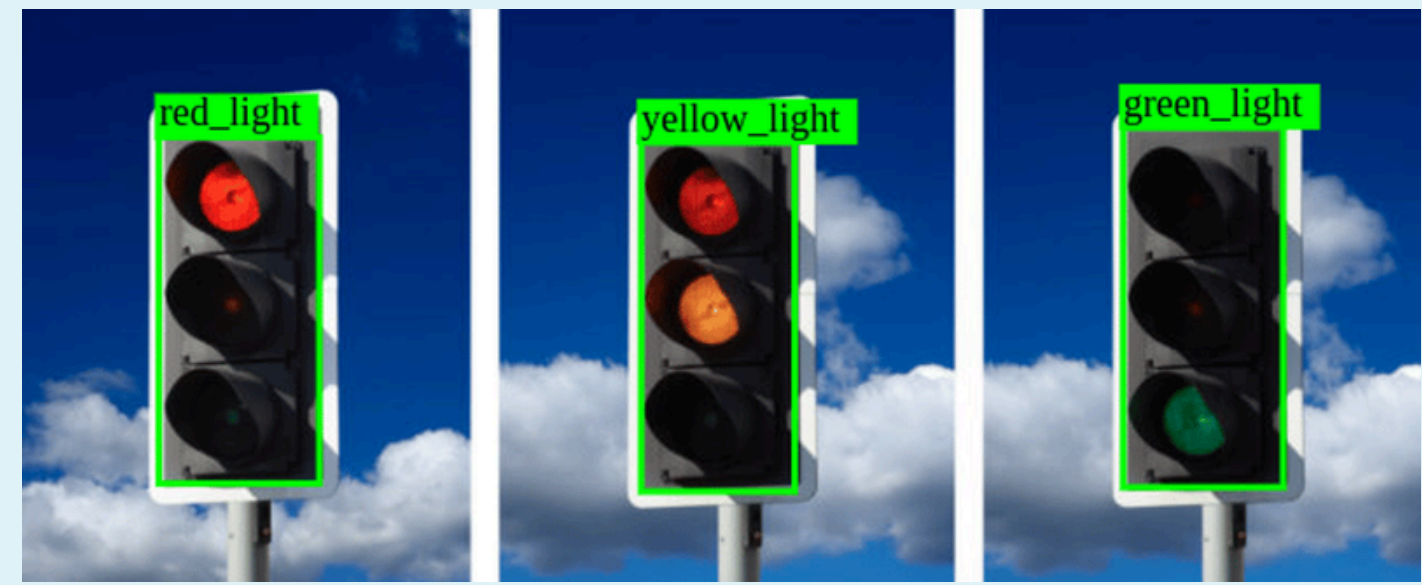
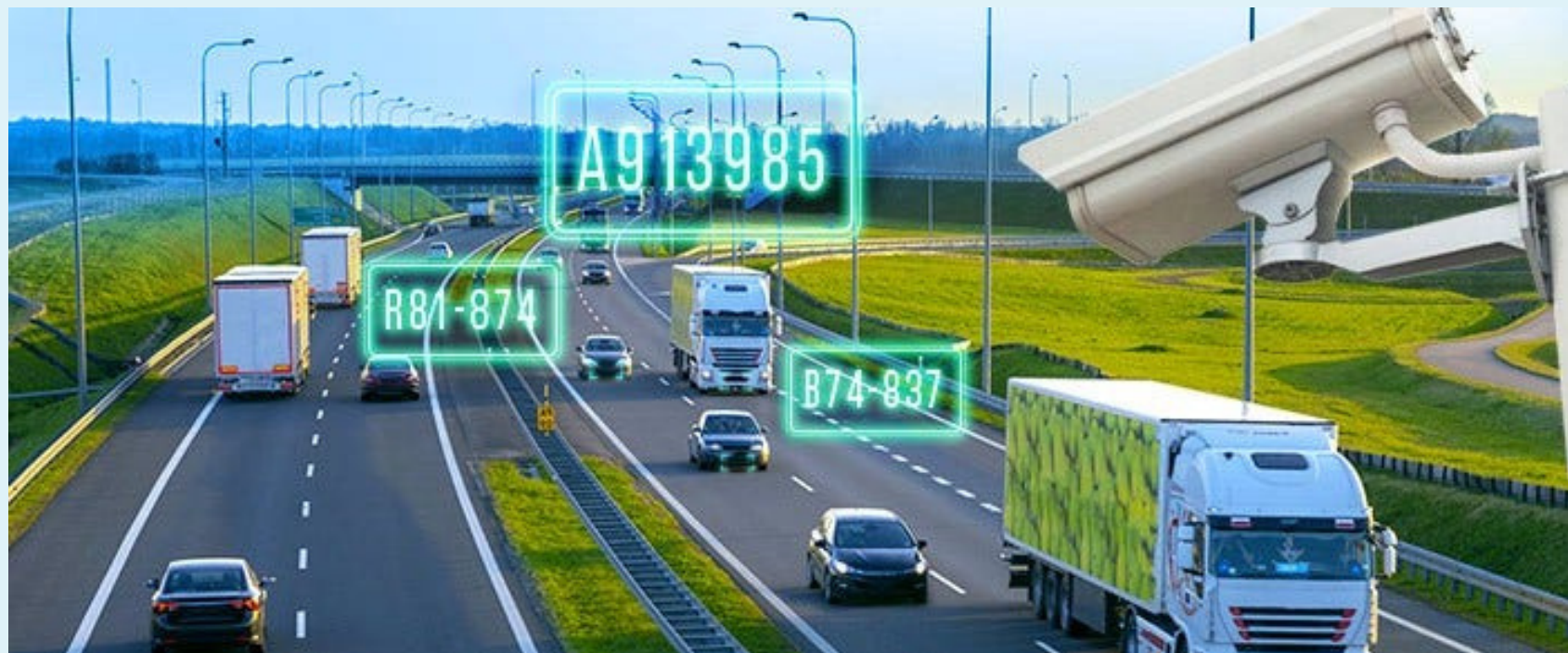
Red Light Detection: Identifies vehicles running red lights in real-time to enforce traffic laws.

Vehicle Detection: Detects and classifies vehicles (cars, trucks, bikes) for monitoring traffic flow and violations.

License Plate Recognition: Reads and records vehicle license plates for identification, tracking, and enforcement.

Violation Logging: Automatically logs detected violations (e.g., speeding, illegal parking) for future reference or legal action.

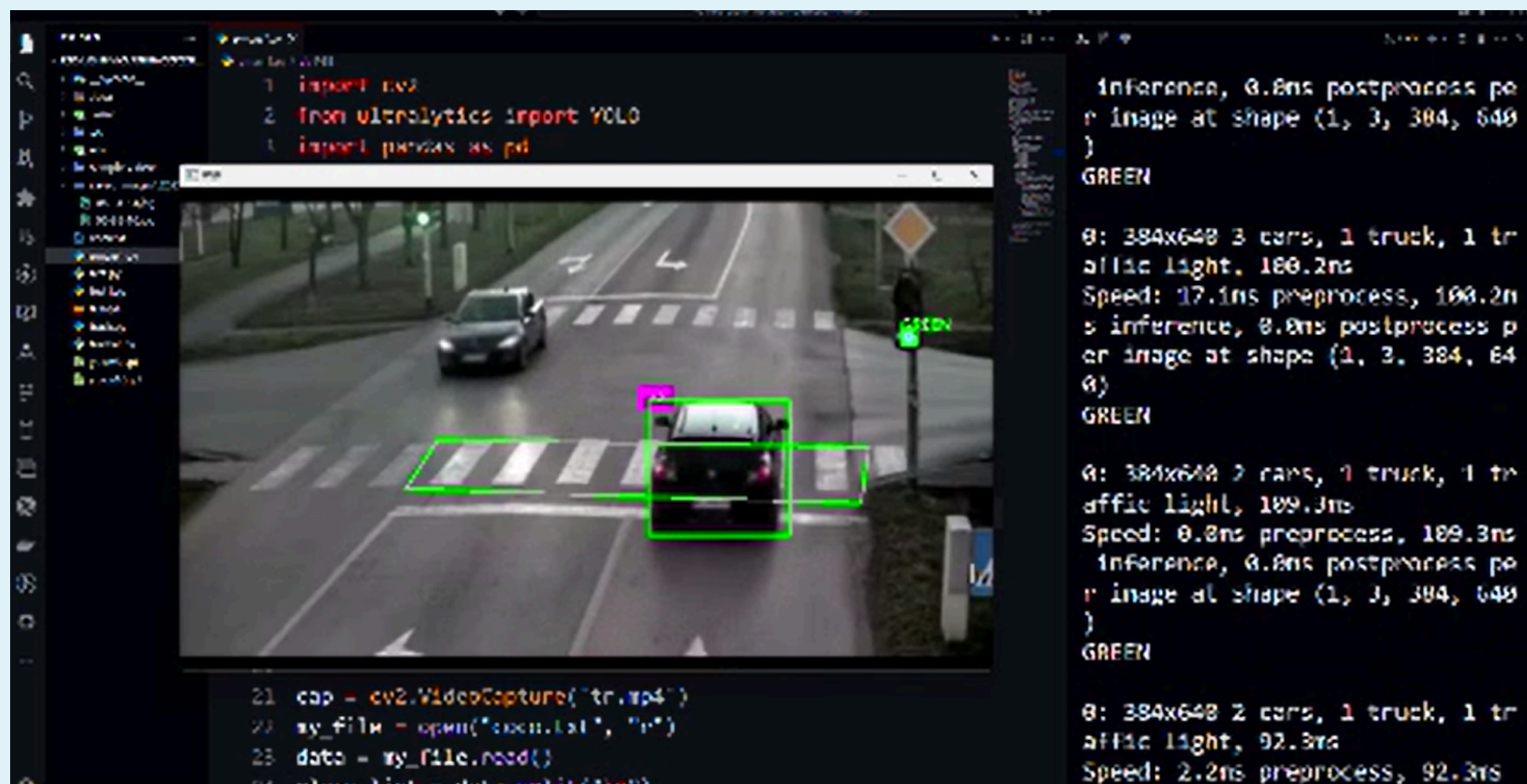
Real-Time Processing: Processes video feeds in real-time to detect and classify objects immediately, ensuring prompt decision-making.



TOOLS & TECHNOLOGIES

1. **OpenCV** : Image processing and object detection.
2. **TensorFlow / PyTorch** : Deep learning for model training.
3. **YOLO / SSD** : Pre-trained models for real-time detection.
4. **CUDA** : GPU acceleration.
5. **Tesseract** : License plate recognition.
6. **Edge Computing** : Real-time processing on devices like Raspberry Pi.
7. **MySQL / MongoDB** : Data storage.
8. **Docker / Kubernetes** : Deployment and scaling.

PROOF OF OUTCOME



ADVANTAGES

- **Improved Safety** - Detects hazards and violations.
- **Real-Time Monitoring** - Instant issue identification.
- **Scalable** - Can be deployed in multiple location.

DISADVANTAGES

- **Privacy Concerns** - Potential issues with surveillance and data privacy.
- **Complex Maintenance** - Requires regular updates and maintenance.
- **High Initial Cost** - Expensive to set up and deploy.

CONCLUSION

In conclusion, the Object Detection System for Traffic Management plays a crucial role in enhancing road safety, optimizing traffic flow, and automating law enforcement in real time. By seamlessly integrating with existing traffic infrastructure, it ensures more efficient management of traffic, contributing to smarter, safer cities. As technology continues to improve, the system's capabilities will expand, further revolutionizing traffic management and making urban environments safer and more efficient.

OUR TEAM

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THANK YOU